

Research synthesis and decision record

Property-management business design | Research cutoff: 15 September 2026

This document explains the evidence, assumptions, alternatives and tests behind the playbook. It provides an inspectable analytical record, not a raw internal reasoning transcript. The detailed source reviews, operating tables, raw public data and calculations accompany it.

1. The question and the research boundary

The question is not how much of today's property-manager job can be replaced. It is **which recurring property outcome a new company can improve enough to earn a sustainable fee, and what division of people, software, authority and physical work makes that possible.**

The work proceeded across the full operating model before narrowing the proposed first offer. It covers ownership and management mandates; marketing and occupancy; physical assets and work; money; obligations and disputes; people, vendors and systems; ordinary operation, changes, failures and exit. Residential, commercial, associations, affordable housing, student/shared housing and hospitality differences are explicit.

The quantitative depth is greatest in US residential rentals. UK commercial standards, English landlord data and US federal/state/local requirements provide useful comparisons and scope checks. This is a systematic broad research corpus, not a claim to have enumerated every supplier, jurisdiction or individual decision in world property markets. Local operating rules still require actual contracts, program details, buildings and current law.

What was examined

- All three supplied attachments, including the complete 13-page strategy review and 12-page McKinsey PDF with exhibits; both linked articles in full.
- An inventory of 38 entries (37 current providers and one acquired product lineage), across PMSs, AI, maintenance, inspection, payments, accounting, deposits, compliance, commercial operations and associations, supported by 75 opened primary-source pages with access/pricing limits.
- Nine cross-industry primary research records with populations, methods, observed results and transfer limits; source tracing for the Forbes statistics.
- The 2024 Census Rental Housing Finance Survey public microdata: 4,425 property records, survey-weighted calculations, codebook, methodology and official verification controls.
- Current Census rental-market statistics, BLS labor context and operator-reported turnover/operating results. These measures retain their different populations and dates.
- A 115-decision inventory across 17 families, with explicit alternatives, authority, execution, dependencies and validation questions. Counts describe the work performed; they do not measure

commercial importance.

No operator interviews, vendor quotes, authenticated PMS trials or live customer experiments were conducted. The attached playbook's private correspondence summaries were considered as supplied context, not independently authenticated as a complete case record. No customer demand or improvement is represented as observed.

2. What survives from the supplied material

The original playbook's strongest insight is that useful software can coexist with a fragmented experience. Its CBS examples distinguish a continuing household from a vacant-unit turn and show that one problem can cross maintenance, accounting, resident-controlled information and a utility. Those are useful distinctions. They do not show how frequently the same problem occurs elsewhere or who would pay to change it.

The strategy review adds commercial urgency but overstates what follows. It calls the selected experience representative, treats limited full-process adoption as a technical frontier, elevates deposit administration before comparing the full business, and supplies prices/margins/revenue milestones without operating measurements. The new playbook retains concurrent research, building and selling while removing those unsupported conclusions.

McKinsey's supplied paper supports connected workflows and explicit attention to who captures value. Its forecast covers real estate, construction and development, not a property-management software market. Its technical-potential diagrams are modeled combined-sector work hours. Its client improvement examples lack the disclosed cohorts and comparisons needed to forecast a new product's effect. Page 4 allows some judgment work to become controlled repeatable work; the simple steps-versus-thoughts slogan should not become a fixed design constraint.

Palantir provides a useful pattern: link relevant facts, evaluation methods, authorized action and the subsequent result. The central Onyx example is fictional. This supports a design principle rather than proof of a particular architecture's ROI. A small operation can apply the pattern with a modest database, calculations and reliable interfaces. [Palantir article](#).

Consequential source corrections

Claim	What inspection established	Decision consequence
8% of firms fully automated a process means a structural automation ceiling	It is a vendor survey adoption measure, alongside 58% reporting AI use; the combined 3,200+ research respondents include managers, owners and renters	Do not infer task capability, 92% addressability or a permanent services niche
8.6% renewal uplift supports residential AI economics	The primary paper concerns willingness to renew in observational office-tenant data	Exclude from residential treatment-effect assumptions
23% retention, 28% training, 85% investor expectation	The inspected source chains did not substantiate these as usable empirical estimates; training also changes measure/denominator between articles	Exclude from demand and savings calculations
Broad read/write API access is available from an AppFolio Plus subscription	Public pricing describes the customer API as read-only; partner access is a separate question	Validate exact operations before committing to a product design
A manager can simply place an outside service under its license	Authority and licensing/exemptions are action- and jurisdiction-specific	Do not turn a service contract into a claimed legal exemption
Colon v. Martin supports deposit remedies	The cited case concerns municipal pre-action examinations	Remove the erroneous case citation; establish applicable statutory rules directly

Sources: [Buildium report](#), [Hu/Kok/Palacios](#), [AppFolio pricing](#), [Virginia exemptions](#), [Colon v. Martin](#). The full audit supplies each intermediary source and the remaining uncertainty.

These errors matter because they alter product selection and economics. A large penalty does not establish incidence, avoidability or willingness to pay. A cited percentage with a different population cannot be used as a shortcut to expected returns.

3. The operating model changes the unit of analysis

The appropriate unit is often an **obligation or outcome that joins several decisions**, not a message, work order, resident or building alone. A unit and an agreement do not have the same lifecycle; several residents can share one agreement, and a physical system can serve several units. One incident can generate several jobs and financial consequences. One job may satisfy only part of a resident's problem.

The detailed map makes seven concurrent lifecycles explicit and separates four flows: physical use/work, money, authority/obligations, and information/commitments. It includes takeover, normal service, renewals, disputes, casualty, insolvency, management change and residual duties after exit.

The decision register is organized around what managers actually choose. It does not assume managers personally hold every right. For each item it asks who may recommend, decide, act and verify; those may be different parties. Sources anchor operating and legal domains; the proposed automation modes are

original hypotheses requiring validation.

What changes across property segments

Segment	What becomes central	Why a single generic workflow fails
Scattered homes	Travel, heterogeneous assets, many owners, outsourced trades	Building-level staffing and equipment assumptions do not hold
Conventional multifamily	Shared systems, onsite teams, frequent leases and service	A unit issue can affect common infrastructure and other occupants
Student/shared housing	Beds, guarantors, changing participants, synchronized dates	A household change is not necessarily a vacant-unit turn
Affordable/program housing	Eligibility, assistance, recertification, program inspections	Standard private-rental terms and decision rights may not apply
Office/retail/industrial	Negotiated obligations, expense recovery, service standards, tenant improvements	A residential rent/deposit model omits major commercial decisions
Associations/co-ops	Boards, voting, reserves, assessments and common property	The payer and decision-maker are collective; owner/tenant roles overlap
Hospitality/specialist use	Short stays, bookings, staffing, operator-specific services	Property and hospitality/care operations must be separated

The map should remain broad even when the product is narrow. It reveals side effects and expansion possibilities; it does not require implementing every relationship in software.

4. Independent public-data analysis

Method

The calculation uses the Census 2024 RHFS public CSV, WEIGHT for properties and WEIGHT × NUMUNITS_R for units. It reproduces official property-size and unit-size verification totals to the published rounding. The included script records the raw-file SHA-256, checks controls and produces the analysis tables.

The financial reference year is 2023. The frame excludes public/transient housing and newly built properties after late 2020. Public-use disclosure modifications affect the unit count; the public-file total is approximately 49.895 million versus 49.722 million in pre-disclosure controls. The former agrees with the public-file verification workbook. These are descriptive point estimates, without newly calculated sampling intervals. See [Census methodology](#), [public CSV](#), and [verification file](#).

Market structure

Property size	Share of represented properties	Share of public-file units
One unit	82.9%	31.5%
2-4 units	14.6%	15.4%
5-24 units	1.7%	8.2%
25-49 units	0.4%	5.4%
50+ units	0.4%	39.4%

Day-to-day manager	Share of represented properties	Share of public-file units
Owner or unpaid agent	53.4%	33.1%
Owner-employed manager	14.7%	22.0%
Management company	21.6%	35.3%
Other	5.4%	2.8%
Not reported	4.9%	6.7%

These calculations show why property counts, unit counts and buying-company counts must stay separate. A manager can aggregate many scattered properties; a large property is not necessarily a large independent buyer. The management-company category does not establish whether the company is affiliated with the owner. Neither table provides a customer list, reachable budget or automatic TAM.

Expenses and missing data

Among valid nonnegative maintenance-and-repair responses, the property-weighted median annual cost per unit is \$1,600 and the unit-weighted median is approximately \$1,175. Only 46.9% of the weighted property population has a valid response for that item. These are conditional descriptive figures, not industry-wide savings potential.

Management-company expense has a valid-response share of only 18.1%. The corresponding medians are \$1,300 and \$720 per unit annually. They cannot be treated as a general quoted management fee. Missing and not-applicable values are excluded separately, never converted to zero. Medians across expense categories have different response populations and cannot be summed into a cost waterfall.

Business inference: maintenance is a material expense and the payer structure is diverse. **Not established:** how much spending is avoidable, how often coordination causes it, or what Handoff could capture. The economic memo and workbook retain these distinctions alongside the raw data.

5. Current competitive reality

Core platforms and specialists now advertise execution, not just records. AppFolio, Yardi, RealPage, MRI and Entrata position agents across operations; EliseAI spans substantial resident and maintenance work. Maintenance specialists extend into cost checks, scheduling, procurement and physical service networks. Financial and deposit specialists cover other proposed entry points.

The inventory establishes **documented coverage**, not actual performance. Recent announcements are distinguished from verified availability. Published pricing is a dated snapshot; quote-only offerings remain quote-only. Supported integration must be checked at the operation and permission level.

This changes the hypothesis from “nobody has built the connective software” to “can we serve a specific customer's operating need better than its feasible alternatives?” The latter can support a business in a competitive category, but requires direct comparison.

Sources and granular observations are in the provider inventory and opened-page audit. Particularly important comparisons include [Property Meld cost support](#), [LeadSimple workflows/pricing](#), [Lula](#), [Lessen](#), and [EliseAI implementation requirements](#). A properly configured existing product is a real competitor, not a control group we are allowed to keep artificially weak.

[Vendoroo](#) is an especially close service comparison because it uses the customer's vendors and markets human support and owner-approval handling. APM Help, OJO and Proper strengthen the financial-operations comparison. Conservice and Second Nature illustrate utility and resident-service alternatives. Their coverage does not prove that any particular deployment solves a customer's problem, but it removes easy claims of category novelty.

6. Why turn readiness is the lead test, and why it can lose

The broad company thesis is to operate a recurring property responsibility with AI and a smaller amount of necessary human coordination. The lead test is vacant-home readiness because it combines an understandable result, observable condition/date, a bounded cohort, several consequential choices and owner-level economic stakes.

It is not objectively established as the best lane. Occupied maintenance is more frequent; takeover projects may have stronger urgency and immediate budgets; financial operations have a recurring established buyer; commercial obligations may support larger fees. Expertise and competitive objections apply to **all** of these, including turns. Physical inspection, contractor management and habitability obligations are not trivial merely because a founder can build the software.

Candidate	Condition that would make it beat turns	Measurement needed
Occupied maintenance	Frequent avoidable repeated visits/approval delays; turns mostly wait on demand	Eligible volume, preventable cost, completion quality and buyer capture
Takeover/onboarding	Partner grows through repeated acquisitions or management assignments	Event cadence, implementation cost, errors prevented and purchase budget

Candidate	Condition that would make it beat turns	Measurement needed
Month-end/financial operations	Controller owns painful recurring work and a capable finance partner is available	Reconciliation burden, error cost, close timing, trust controls and alternative quote
Commercial obligations	Accessible specialist/operator with collectible contractually supported value	Actual lease sample, repeatable obligations, cycle length and net recovery
Existing-stack implementation	Needed functions already work once configured; no ongoing delivery gap	Setup effort, useful transfer and whether recurring service is actually necessary

The first discovery week therefore compares real work across domains before contracting a turn pilot. This preserves a clear lead hypothesis while allowing evidence to select another business.

The strongest counterthesis

Turn coordination could be a low-volume service whose apparent value comes from rent gains that never materialize. Skilled trades, capital or rental demand may dominate the timeline. An existing coordinator with standing approvals may perform just as well. Native vendors and operated services may already cover the relevant segment at lower cost. If so, AI adds another layer of supervision rather than removes work.

The response is not a more elaborate system. Test the actual critical path, compare the best alternatives, and require enough practical control to improve the outcome. Do not guarantee a date while lacking access, vendors, funding or authority. A pilot can initially promise specific controllable service actions and report the complete outcome; any later performance guarantee should match demonstrated control and contractual allocation.

7. Economics that can falsify the recommendation

The live workbook uses one clearly labeled **illustrative turn scenario**. None of its savings or prices is an observed customer result.

Input	Assumption
Portfolio and turnover	1,000 units; 35% annual turns; all eligible
Monthly case volume	29.17 turns
Net manager time released	60 minutes/turn at \$45/hour
Cash realization of released time	50%
Additional collected rent	3 days/turn at \$2,000 monthly rent, using 30 days
Avoided repair expense	\$0
Service price	\$100/turn; no recurring base fee

Input	Assumption
Handoff human work	35 minutes/turn, including assumed exceptions
Handoff labor and variable tools	\$35/hour; \$2/turn
Account support and setup allocation	\$400/month plus \$1,000 setup spread over 12 months

Monthly result	Integrated owner-operator	Third-party fee manager at 8%
Benefit captured before service price	\$6,490	\$1,123
Service price	\$2,917	\$2,917
Buyer net benefit	\$3,573	-\$1,794

Handoff's modeled monthly delivery cost is \$1,137 and contribution is \$1,780, or 61.0%, before acquisition, central overhead and product development. The direct break-even price is \$38.99 per turn; a 60% contribution target under these assumptions requires \$97.47. These are algebraic thresholds, not market prices or validated margins.

The crucial downside: with no additional collected-rent days, the integrated buyer loses approximately **\$2,260 a month** at the same price. With no repair savings, the integrated buyer needs about **1.16 genuinely additional paid days per turn** to break even. A fee manager capturing only 8% of incremental rent needs approximately **14.53 days** under the same assumptions. This strongly favors an owner-funded budget or a different source of manager value; it does not prove that an owner will buy.

The separate general maintenance-work illustration in the economic memo is deliberately a different population: 210 eligible items/month rather than 29 turns, different handling/pricing, and explicit hypothetical expense savings. It is a comparison of business mechanics, not a second estimate of the same service. The workbook and script use identical assumptions for the turn example.

Frequency and measurement

Invitation Homes' 22.8% FY2025 same-store turnover implies 19 monthly turns per hypothetical 1,000 homes. AvalonBay's 37.2% annualized H1 2026 turnover implies 31. These portfolios differ; the figures are not a statistically estimated industry range. They show why a small company's short pilot may have too few turns. [Invitation Homes](#), [AvalonBay](#).

A sample of 20-40 cases is useful for discovering workflow and unit-cost issues. It cannot establish a stable small causal effect or rare-failure safety. As a mathematical illustration, detecting a two-day average change with five-day standard deviation at conventional 80% power and 5% two-sided significance requires about 98 cases per group before clustering or other complications. The field guide separates feasibility, economics and later causal evaluation.

8. Cross-domain lessons that earn their place

The research corpus includes customer support, consulting, product development, office work, software maintenance, economy-wide labor adoption, clinical reasoning and building analytics. Positive and negative results both matter; older model studies are not permanent capability ceilings.

Transferable lesson	Application here	What is not justified
AI can spread useful operating knowledge and reduce novice ramp time	Reuse successful diagnosis questions and well-founded decisions	Importing support-center productivity percentages into PM staffing
Capabilities vary by task; human-plus-AI can fail	Test repair options, permissions, invoice matching and escalation separately	A universal approval checkbox as proof of safety
Individual time savings need not change organizational output	Redesign approvals, staffing and dependencies, not only drafting	Counting faster email as payroll savings
Better analysis can cross departmental boundaries	Prepare a combined cost, timing and resident-impact decision	Assuming the model has physical expertise or contractual authority
Physical-system savings require action and verification	Link observed conditions to feasible repair and outcome checks	Claiming sensor-based energy savings for homes without those systems
Selection and self-report can mislead	Include failures, refused cases, rework and actual clocks	A successful demo or enthusiastic survey as proof of impact

The primary-study table in the corpus records methods and limitations. It supports these evaluation choices, not the financial inputs. The most useful inheritance from Owen's projects is disciplined comparison and execution: reliable calculations, explicit applicability, source-aware interpretation, practical integration, and testing with real cases. The company should not inherit their vocabulary or architectural scope.

9. Recommendation changes and remaining tests

Question	Evidence or analysis	Resulting decision	What would change it
Is software missing?	Broad current provider coverage and implementation documentation	Compete against feasible existing execution, not absent technology	A repeatable task with no adequate supported product/service
Is the residual work inherently human?	Adoption does not establish capability; task-level experimental variation	Separate ability, permission, observability, cost and preference	Measured model limits in the selected operating context
Should deposits lead?	Prior ranking not based on broad comparison; legal/source errors and active competition	Reopen all lanes; deposits remain one candidate	Actual volume, costly preventable failures and funded delegation
Who should buy?	Owner/manager value split and live economics	Prefer owner-operator or explicit owner-funded budget	Verified staffing/capacity gain makes fee-manager purchase work
What should be sold?	Feature competition; need for actual completion	A bounded operating result, with software behind it	Customers already supply execution and only need a tool
Why turns first?	Understandable result, connected choices, measurable cohort	Provisional lead test with broad first-week comparison	Demand-limited vacancy, low volume, insufficient control or better adjacent economics
How much should be built?	Integration/authority can dominate implementation	Small working slice through real permitted actions	Repeated customer use establishes need for broader shared capability
How should it scale?	Exception cost and repeated setup threaten service economics	Reproduce delivery with a second partner before expansion	Service cannot repeat profitably; remain a priced project business or stop

10. Evidence still missing

The highest-value next evidence is private operating evidence obtained with permission: a consecutive case history, actual approval limits, installed modules and configuration, required read/write permissions, staff time, contractor performance, readiness and collection dates, and a real purchasing decision.

Public research cannot determine those facts. Specifically unresolved are: target customer willingness to delegate and pay; actual preventable delay; local supplier availability; integration costs and commercial access; role-specific labor costs; callbacks and quality; founder/operator execution capability; and customer acquisition cost. The operating map identifies what to ask for without presuming the answer.

The property business as it actually operates

Purpose. This is an original operating and decision model, informed by the primary sources listed in `operating_sources.csv`. It is a broad working map for discovering a business, not a claim that one process fits every property. The companion `decision_inventory.csv` contains 115 distinct decision prompts across 17 families, with triggers, inputs, choices, authority, execution, outcomes, constraints, automation hypotheses, dependencies, and questions for operator validation.

Coverage boundary. It covers conventional rental housing, scattered homes, multifamily, student and affordable housing, associations, commercial space, and hospitality variants, including normal operation, stress, and exit. A locally exhaustive implementation also needs the actual management agreement, leases, physical systems, loan documents, insurance, program rules, governing documents, and applicable current law. The inventory is not a measured distribution of time, case frequency, avoidable cost, or automation feasibility. Its source IDs identify supporting operating domains; they do not imply that a source endorses our product hypotheses.

1. What the business is for

People and businesses need useful space. Somebody owns that space and must pay to preserve it. Other parties supply labor, capital, utilities, protection, and rules. Property management makes the resulting promises work day after day: people can use what they were promised; problems are resolved; the right money reaches the right party; the property remains usable; the owner can make informed choices.

There are **three different businesses** that often appear in one software account:

Business	Economic purpose	Typical decision owner	What must not be conflated
Property ownership/investment	Provide useful space and earn an acceptable return within obligations	Owner, investor, asset manager, board	Property value and collected rent are not management-company revenue
Property management	Deliver agreed operations profitably and retain clients	Management principal, regional manager, site manager	Owner savings do not automatically become manager savings or willingness to pay
Occupant's household or business	Live, work, trade, produce, receive care, or stay temporarily	Household members, tenant company, guest, specialist operator	Occupants are people with rights and purposes, not merely sources of rent or service demand

A vertically integrated owner-manager can align some tradeoffs internally. A third-party manager must negotiate them. An association manager serves collective governance; the board and members retain decisions defined by law and governing documents. A building manager, investment asset manager,

facilities contractor, hotel operator, and licensed healthcare operator do not hold interchangeable authority.

California's official property-management reference confirms the breadth of the role: agency agreements, owner and tenant duties, leasing, maintenance, accounting, staffing, purchasing, and reporting. It is used here as a scope check, not a source for current prices or a universal legal rule. [001 - DRE](#)

2. All the parties and what connects them

One person or firm can fill multiple rows; a row cannot disappear merely because roles share an employer.

Party	What it supplies or receives	Decisions/rights it usually holds	Main connection to the manager
Beneficial owner; property-owning entity; investor; fund	Equity, objective, risk capital; distributions and property value	Investment strategy and reserved approvals	Mandate, budgets, approvals, funding, reporting
Asset manager; owner representative; family office	Portfolio direction and capital allocation	Delegated owner decisions	Operating plan, exceptions, forecasts, transaction preparation
Association/co-op board; committees; voting members	Collective authority, assessments, shared-property decisions	Reserved votes, budgets, rules, capital decisions	Meeting materials, approvals, enforcement, reserves, records
Management-company principal; licensed broker where applicable	Contractual operating service and supervision	Client acceptance, staffing, delegated actions	Scope, economics, supervision, escalation, liability
Regional/site/property manager	Day-to-day coordination and decisions	Authority granted by mandate, policy, law	Owns execution across people and systems
Leasing/brokerage; marketing; reservations staff	Demand, tours, negotiation, booking	Delegated offers and representations	Availability, pricing, approved terms, qualification
Bookkeeper; controller; treasury; payroll; CPA	Accurate books, client funds, payments, filings	Financial controls and approved releases	Bill/receipt matching, cash availability, distributions, close
Prospective occupant; applicant; tenant; resident; guest	Demand, payment, information; receives use and service	Application, agreement, notice, complaint, lawful remedies	Inquiry through move-out and later disputes

Party	What it supplies or receives	Decisions/rights it usually holds	Main connection to the manager
Roommates; household members; guarantor; emergency contact; representative	Shared use, guarantee, consent, emergency support	Rights differ from named payer or leaseholder	Occupancy, access, changes, lawful disclosure
Housing authority; subsidy administrator; tax-credit allocator; service coordinator	Assistance, eligibility rules, monitoring, resident support	Program approval and oversight within program	Certification, assistance payment, inspection, transfer, reporting
Maintenance employee; building engineer; cleaner; concierge; security	Physical service and observations	Role-specific operational and technical authority	Work, access, condition, safety, shift coverage
Trade contractor; general contractor; subcontractor; supplier	Skilled work, parts, capacity, warranties	Methods within qualification and contract	Scope, bid, dispatch, progress, change, completion, invoice
Architect; engineer; surveyor; inspector; environmental specialist	Qualified assessment or certification	Professional conclusions within scope	Design, safety, condition, permits, reserve studies
Utility; telecom; waste; meter provider; district service	Essential infrastructure and measured usage	Service/network control and tariff administration	Account transfer, interruption, repair, billing, capacity
Bank; payment processor; card network; escrow provider	Settlement, account custody, payment events	Payment controls, returns, holds	Receipts, releases, reconciliations, fraud responses
Lender; servicer; lender's inspector	Debt and restricted reserves	Covenants, consents, draws, cash management	Reporting, reserve release, repairs, insurance, default
Insurance broker; insurer; adjuster; restoration firm	Risk transfer, claims determination, recovery	Policy coverage and claim approval	Risk changes, renewals, incident notice, mitigation, payment
Lawyer; mediator; court; enforcement officer; receiver; trustee	Advice, dispute resolution, enforceable orders	Legal representation, adjudication, lawful enforcement/control	Notices, claims, possession, insolvency, transfer of authority

Party	What it supplies or receives	Decisions/rights it usually holds	Main connection to the manager
Building/fire/health/environment/rental regulator; tax authority	Public standards, permits, inspections, taxes	Enforcement and statutory approvals	Registration, filings, inspections, correction, appeals
Emergency responders; neighboring owners; adjacent occupants	Immediate response and shared exposure	Emergency powers; independent rights	Hazards, access, common systems, incident communication
PMS; CRM; access-control; accounting; screening; communications; AI vendors	Records, calculations, workflows, interfaces	Only permissions delegated to software/provider	Data, actions, audit history, outages, export, switching
Buyer; seller; developer; outgoing/incoming manager	Asset or responsibility transfer	Transaction-specific rights	Diligence, assignments, opening balances, keys, unfinished obligations

The important separation is **who wants the result, who pays, who can authorize it, who does it, and who can verify it**. These are often five different parties. A better recommendation alone does not join them.

3. Four flows run through every operation

Flow	Ordinary path	Typical break	Product implication
Physical use and work	Space becomes usable; occupant takes possession; equipment is serviced; defects repaired; space returned	Access denied; diagnosis wrong; part absent; contractor unavailable; work marked complete prematurely	Digital progress must correspond to a real change in the property
Money	Occupant/subsidy pays; bank settles; ledger allocates; operating bills and debt paid; required reserves retained; owner receives available cash; manager earns agreed fees	Returned payment; wrong owner ledger; restricted cash mistaken for spendable cash; duplicate invoice; unapproved cost	The account with cash is not necessarily the party entitled to spend it
Authority and obligations	Law and contracts establish rights; owner/board delegates; PM approves; qualified parties execute; regulator/court may supersede	Owner absent; unclear scope; unsigned amendment; expired power; court hold; contradictory instruction	Action rights need actual scope, limit, effective date, and change handling
Information and commitments	A report enters; facts are checked; a decision made; someone commits; updates and completion return	Multiple inboxes; stale status; silent waiting; disputed facts; undocumented promise	Track who will do what by when and whether the promised result happened

The sequence is not always linear. Emergency mitigation can precede cost allocation. A vendor can complete work before an invoice arrives. A lease can be signed before the unit is ready. A payment can appear before it finally settles. A property can sell while a claim remains open. The model must allow these combinations rather than overwrite them with a single status.

For example, California's trust reference distinguishes bank records from beneficiary records; reconciling them is essential to knowing whose money is present. That distinction is the useful design lesson. It is not an instruction to apply California's precise rules everywhere. [002 - DRE trust funds](#)

4. Concurrent lifecycles and their states

These lifecycles coexist. A building does not move from leasing to maintenance and stop doing the other work.

Lifecycle	Main states	Exit or recurrence	Important simultaneous condition
Ownership and management mandate	Prospect; diligence; agreement pending; signed; takeover; active; changed scope; termination pending; handover; residual obligations; archived	Renew, sell, change manager, restructure, dissolve	Legal owner, operating authority, and bank signer may change on different dates
Physical property and equipment	Planned/acquired; delivery/commissioning; usable; degraded; restricted; failed; temporary remedy; repair; verified usable; renovation; decommissioned	Repeated operation and renewal; redevelopment; demolition	Occupied space can be partly unusable; usable space can be unavailable to lease
Demand and occupancy rights	Available/coming available; prospect; tour; application; waitlist; offered; accepted; signed; pre-possession; occupied; amended; renewal; notice; holdover/disputed possession; returned; final account; residual claim	Renew or relet; cancellation; transfer; lawful termination	Marketing availability, legal possession, and physical readiness are distinct
Service request or planned work	Reported/due; identity located; urgency assessed; facts missing; scoped; responsibility identified; authorized; funded; assigned; scheduled; in progress; blocked; completion claimed; verified; billed; paid; reopened/closed	Callback; warranty; recurring service; capital escalation	One incident can require multiple jobs; one job can serve multiple units
Money and accounting	Obligation estimated; charge/invoice issued; disputed; approved; payment initiated; pending settlement; settled; allocated; reconciled; reported; refunded/adjusted/written off	Recurring rent, service invoices, assessments, taxes	Cash, receivable, expense, deposit liability, and owner distribution are different things
Complaint, legal matter, claim, or emergency	Detected; protected/contained; assessed; notified; investigated; remedy proposed; contested; approved/ordered; carried out; recovery/settlement; appeal/reopened; retained	Recurrent condition, further loss, new claim	Ordinary collection or access workflows may be suspended while essential service continues

Lifecycle	Main states	Exit or recurrence	Important simultaneous condition
People, vendors, systems and capacity	Needed; selected; verified; granted access; trained; active; reassigned; unavailable; suspended; replaced; access revoked	Renewal, hiring, turnover, outage recovery	A valid contract does not mean capacity is available today

Do not infer a finished result from an intermediate state. “Notice received” is not “possession returned.” “Paid” is not “settled.” “Invoice approved” is not “work verified.” “Vendor assigned” is not “appointment confirmed.” “Manager offboarded” is not “all duties extinguished.”

5. Transitions, exceptions, and reversals

The table identifies consequential joins. It does not prescribe a universal order for every jurisdiction.

Transition	Conditions to establish	Exception or reversal	Consequence for connected work
Agreement signed → active management	Effective mandate; owner identity; funding; access; emergency contacts	Closing slips; authority contested; essential records absent	Conditional takeover; preserve emergency coverage; do not assume all powers started
Old manager → new manager	Funds/liabilities reconcile; records and keys received; open matters assigned	Missing deposits; disputed invoices; old credentials active	Separate unknown opening balances; revoke access; assign surviving duties
Future vacancy → marketable offer	Forecast date and permitted marketing; truthful attributes	Notice withdrawn; holdover; works delayed	Update prospects, commitments, staffing, and financial forecast
Applicant → accepted tenant	Permitted screening; actual decision authority; program fit	Report error; accommodation; queue priority; adverse action	Correct decision, provide applicable notice, preserve dispute path
Accepted → signed → possession	Operative agreement; required disclosures; funds conditions; safe readiness; legal availability	Signature fails; payment returns; current occupant remains; unit unsafe	Delay/alternative/credit as lawful; do not issue misleading access or readiness
Report → repair response	Correct location; severity; safe next step	Emergency; duplicate report; ambiguous unit; multiple affected systems	Immediate protection; merge facts without losing distinct affected occupants
Scope → approved/funded work	Correct responsibility; sufficient authority and cash	Owner silent; estimate exceeds cap; mandatory urgent duty	Defined emergency process; specific approval; owner funding; lawful mitigation

Transition	Conditions to establish	Exception or reversal	Consequence for connected work
Assigned → attended → repaired	Confirmed provider; parts; lawful access; adequate diagnosis	No-show; no access; hidden damage; parts unavailable	Replan with named blocker and next commitment; update occupant
Completion claimed → verified	Original problem tested; competent assessment; unresolved issues recorded	Callback; resident disagreement; new defect	Reopen linked work; preserve history; avoid duplicate-charge assumptions
Invoice → payment	Approved scope; delivered work; payee verified; correct entity and funds	Duplicate; changed bank details; disputed scope; lien issue	Hold appropriate payment; verify independently; pay undisputed amount only when authorized
Receipt → settled available cash	Bank confirmation; allocation; reserves and restrictions recognized	Return; chargeback; wrong payer; unidentified receipt	Reverse correct entry; reassess arrears and distributions; no silent overwrite
Arrears → enforcement	Accurate debt; required notices; authority; no applicable hold	Repair dispute; assistance; bankruptcy; protected circumstance	Pause prohibited actions; counsel; settlement; corrected balance; essential repairs continue
Notice → possession returned	Actual surrender or lawful enforcement; belongings/access addressed	Abandonment uncertain; death/probate; holdover	Counsel and authorized process; delay turn and incoming possession
Vacant → ready → occupied	Sequenced work; inspection; utilities; legally usable space	Failed final check; damage after inspection; new tenant cancels	Reopen readiness; revise availability and commercial commitments
Inspection → compliant	Specific deficiency remedied; required acceptance/certification	Finding contested; remedial work insufficient; deadline missed	Appeal/correction as available; exposure remains until resolved
Casualty → safe reoccupation	Qualified safety finding; restoration; service; required permission	Hidden damage; claim denied; funding insufficient; tenant displacement	Separate recovery negotiation from urgent safety; alternative accommodation and funding plan
Solvent owner → receiver/bankruptcy control	Court order; debtor identity; authority; restricted funds	Changing orders; partial asset coverage; stay relief	Replace approval path; hold affected collection/payments; continue permitted essential operation

Transition	Conditions to establish	Exception or reversal	Consequence for connected work
Operating asset → sale/redevelopment	Transaction; assignment; notices; cash/deposits; tenant rights; work continuity	Sale fails; loan consent absent; possession schedule infeasible	Continue current obligations; reverse provisional changes; retain verified records
Case closed → reopened	New fact; reversal; recurrence; dispute; appeal	Prior record deleted or overwritten	Preserve old decision, record correction and downstream changes

These reversals are not rare technical edge cases to append later. Whether they are frequent enough to justify a product must be measured, but they belong in the operating model from the start.

6. How the property manager makes a decision

A useful decision description asks: **What changed? What do we actually know? What needs to happen? Who can decide? Who will do it? What will count as done? What happens if the plan fails?**

The manager does not personally decide everything. They can originate a recommendation, exercise delegated choice, request someone else's approval, coordinate execution, and verify the result. The same person may switch among these roles in a single incident. A viable product must know which role it is replacing or supporting.

Decision family	Inventory IDs	Main dependency
Management mandate	D001-D007	Staff capacity; vendor coverage; available records
Takeover and operational setup	D008-D014	Mandate; funding; access
Portfolio planning and resource allocation	D015-D020	Budget; leasing; capital
Demand and leasing pipeline	D021-D026	Turn plan; rent approval; inspections
Applications and eligibility	D027-D032	Published criteria; source access
Lease and move-in	D033-D037	Screening; pricing; readiness
Occupied-property service	D038-D044	Occupancy identity; responsibility map
Rent; receivables; arrears	D045-D050	Lease terms; meter; resident change

Decision family	Inventory IDs	Main dependency
Reactive repairs and incidents	D051-D061	Access; contacts; property equipment
Preventive work; capital; energy	D062-D068	Asset inventory; access; vendor availability
Procurement and vendor relationships	D069-D073	Needed trade; service geography
Accounting; cash; reporting	D074-D081	Contract; service completion; property identity
Renewal; vacancy; turn	D082-D088	Pricing; resident issues; unit plans
Legal; insurance; casualty; continuity	D089-D094	Repairs; resident history; legal hold
Management-company people; systems; commercial health	D095-D100	Staff capacity; risk; client scope
Asset sale; refinancing; redevelopment; exit	D101-D104	Closed books; physical backlog; legal matters
Segment-specific additional decisions	D105-D115	Mandate; budget; reserve plan

The CSV's `what_to_validate_with_operator` column is important: it converts a desk-research map into a fieldwork instrument. Ask an operator to walk through completed, delayed, reopened, and disputed examples. Count omissions and rewrite the map around the actual work.

7. Segment differences that change the business

Segment	Operating distinctions	Buyer/authority differences	Product consequence
Self-managed small landlord	Few properties; owner does leasing, approval, bookkeeping and vendor selection; sporadic work	User, payer and owner often one person	Simple low-cost or event-based service; less coordination volume per account
Third-party scattered single-family/small multifamily	Travel; different owners and standards; vendor coverage; separate reserves; heterogeneous homes	PM buys tools but owner may get most property-level benefit	Owner authority and client-level economics can dominate software sophistication
Large institutional single-family	Geographic clusters; centralized staff; standardized policies; remote field execution	More consistent delegated authority and procurement	Route density, repeatable service and integration can improve economics
Conventional multifamily	Shared systems; on-site team; many residents affected by one incident; continuous leasing	Owner/AM and regional/site teams divide rights	Building-wide cause and impact matter alongside unit tickets
Student/shared housing/co-living	Bed and room can be separate saleable units; guarantors; roommates; concentrated turns	Student/household/guarantor rights differ; operator policies matter	Lease identity is not room occupancy; seasonal capacity and allocation are central
Affordable/subsidized/restricted housing	Program eligibility; waitlists; recertification; rent/subsidy split; inspections; agency contact	PHA/contract administrator/allocator may hold key decisions	Program-specific state and dates; never one generic affordable-housing rule set
HOA/condominium/co-op	Common property; assessments; reserves; votes; rule enforcement; architectural requests	Board/member governance; manager implements delegated authority	Approval validity and common/private responsibility matter more than lease funnel
Office	Tenant buildouts; lease abstracts; shared services; after-hours HVAC; expense recoveries	Tenant companies and brokers; negotiated long leases	Clause-specific charges, operating hours and fit-out delivery
Retail	Co-tenancy; permitted use; exclusivity; sales reporting; percentage rent; shared customer areas	Anchor and other tenants' rights interact	One tenant's condition can trigger another lease's economics
Industrial/logistics	Loading; power; fire systems; permitted processes; environmental exposure; roof/slab responsibility	Tenant operates its own production/logistics business	Technical scope, downtime and responsibility split; standard residential repair triage insufficient

Segment	Operating distinctions	Buyer/authority differences	Product consequence
Mixed-use	Residential, commercial and common elements share systems	Multiple contracts and governance layers	Allocate physical impact and cost across distinct legal rights
Hospitality/short-stay	Nightly inventory; channels; housekeeping; guest recovery; lodging taxes; brand/OTA conditions	Operator may control guest service and pricing; owner retains capital	Different tempo, inventory granularity and payment/tenancy rules; separate initial market
Manufactured housing; parking; storage	Land/site/space rights may differ from building ownership; vehicles or goods rather than a household unit	Specialized occupancy and lien/possession rules	Same operating families with different objects and legal consequences
Senior housing; healthcare; data centers; specialist industrial	Building operations coexist with care, clinical, critical-IT or process operations	Licensed/qualified operators hold specialized responsibilities	Distinguish real-estate service from regulated operating business; partner for specialization

Affordable-housing and association cases demonstrate why “all property managers” is not one implementable customer segment. The HUD handbook shows the program-specific selection structure; it is historical and must be supplemented with current program rules before deployment. Florida's condominium statute joins qualified physical reserve assessment to funding and governance. [O07 - HUD](#); [O15 - Florida](#)

Commercial service charges similarly depend on what individual leases permit, how services are procured and allocated, and how actual expenditure is reconciled. RICS is a useful commercial operating reference; its standard is not a statement of US law. [O12 - RICS](#)

8. Representative legal and institutional constraints change the design

The implication is not “legal work cannot be automated.” It is that automation needs the actual rule, relevant facts, valid authority, and a correction path.

Example	Source-supported fact	General operating/design lesson
US tenant screening	A condition such as extra deposit or co-signer can be adverse action when a consumer report contributes to it; notices are not only for outright rejection	Preserve actual decision inputs and trigger the right downstream action; a score alone is insufficient O03
Lead disclosure	Most pre-1978 housing requires specified information before lease signing	Building facts and action order matter; a later document upload may not cure a missed step O05

Example	Source-supported fact	General operating/design lesson
NYC essential services	Heat obligations depend on season/time/temperature; hot-water obligations apply year-round	Urgency cannot be determined by generic keyword alone; property and live conditions matter O08
California possession and deposits	Official guidance prohibits self-help lockout and distinguishes lawful deductions from ordinary wear	Payment status cannot directly trigger lock changes; restoring space and allocating repair cost are separate decisions O13
US algorithmic rental pricing	DOJ's 2025 announcement addresses sharing of competitively sensitive information in a proposed RealPage settlement	Do not assume data pooling or coordinated optimization is permissible merely because technically feasible; examine actual data and market conduct O04
Bankruptcy	Automatic stay and cash-collateral restrictions can change permissible collection and spending; exceptions and orders matter	Authority must change when circumstances change; old automated instructions must be interruptible O10
Hazardous-energy maintenance	OSHA's standard regulates energy isolation during covered servicing	A digital agent can coordinate work; qualified physical safety procedures still have to occur O09
Fair housing	Protected rights apply across housing transactions, not only initial applications	Pricing, service, access, renewal and enforcement all need appropriate treatment; changing the interface does not remove the obligation O14

These are architectural examples, not a legal product specification. Laws, regulations, court orders, agency guidance, professional standards, contracts, and owner preferences have different force. They should not be merged into an undifferentiated “policy” without identifying the source and who may change it.

9. What this map suggests about opportunities

The following are **inferences to test**, not market facts established by the sources.

- 1. Waiting can be more expensive than doing.** Many jobs stall between a known need and authorized, funded, scheduled, physically completed action. Measure elapsed time and its cause as well as staff minutes. A language model that writes faster does not solve lack of a plumber or an owner refusing funds.
- 2. The most valuable unit may be an outcome across several tasks.** “Ready for the promised move-in date,” “repair actually works,” or “correct owner funds available after close” can align product scope with what operators already recognize. It need not require replacing their systems.
- 3. Information can change the process rather than decorate it.** Good diagnosis can eliminate a visit; permission can be agreed in advance; a vendor can receive the right parts before dispatch; jobs can be combined geographically; responsibility can be resolved without withholding urgent service.

4. **Authority is often a redesign opportunity.** A human approval may exist because nobody has written the owner's allowable choices clearly. Pre-authorized classes, spending limits, and escalation conditions can remove delay. This is different from assuming every approval is legally mandatory or all approvals are waste.

5. **Physical observations deserve attention.** The first reliable condition check can prevent downstream disagreement in repair, turnover, billing, and claims. Its value must exceed collection burden; indiscriminate photography and forms are not a product advantage.

6. **Financial benefit follows the contract.** Lower property vacancy may benefit an owner much more than a percentage-fee manager. The paying customer must capture enough value, or the commercial arrangement must change.

7. **Exception handling is not inherently forever-human.** Some exceptions are missing facts, bad records, or unstated policy. Others require technical expertise, negotiation, physical work, or reserved authority. Classify the actual reason, then test whether it can be removed, standardized, delegated, or supported.

8. **Recurrence creates learning only if outcome and cause are known.** A callback may indicate poor work, wrong diagnosis, occupant behavior, or a new defect. Counting "tickets closed" can reward premature closure; learning requires enough attribution to improve future choices.

9. **Human service can be the right delivery model.** A company can sell reliable completion and use software internally before selling a standalone product. The service must still earn sustainable margins and avoid becoming bespoke coordination for every client.

10. **Breadth is a discovery tool, not an implementation commitment.** The map identifies possible markets. A first business should specify one paying customer, one consequential result, one operating setting, one authority model, and a credible way to verify delivery.

DOE's operations guidance supports treating maintenance as a system of interacting functions that affects reliability and efficiency. It does not establish a particular savings percentage for private rental properties or prove any proposed company will capture it. [O06 - DOE](#)

10. How to validate completeness with an operator

Use actual cases, not only interviews about preferred workflows. For each candidate customer obtain, with permission, a small set spanning ordinary work, delayed work, disputed work, repeat failure, emergency, takeover and exit. Reconstruct timestamps and actions from their existing records; distinguish what the records prove from what staff remember.

For every case ask: Who wanted what? Who was entitled to decide? What was missing? What changed physically? What money moved? Who waited for whom? What did completion mean? What later reversed it? What did the manager earn or lose? What did the occupant experience?

Map each observation to the decision register. Add missing decision variants; delete inapplicable ones from the customer-specific version; keep the broad market map intact. If the work cannot be sampled, outcomes cannot be verified, authority cannot be granted, or the beneficiary will not pay, that is a business finding - not an invitation to build a more elaborate architecture.